

Preface

In September 2017, I began working as a violin teacher, primarily teaching children aged 6–9. Although my background in STEM and formal violin training enabled me to identify essential instructional content and communicate it effectively, my lack of prior teaching experience and unfamiliarity with early childhood education represented the greatest challenges. To address these limitations, I decided to apply the statistical knowledge I had acquired during my university studies to systematically monitor and analyze students’ learning progress. By examining these data, I sought to identify instructional strategies that could both enhance learning outcomes and foster an enjoyable learning experience.

The data analyzed in this study were collected between September 1, 2017, and May 13, 2020, covering approximately two years and eight months. The study demonstrates how quantitative observations correspond to real-world learning situations and illustrates how data can be used to address specific learning challenges. To protect students’ privacy, all participants were anonymized using identification codes rather than their real names. Moreover, because every learner is unique, this study does not advocate a universally effective teaching approach. While addressing students’ learning problems remains the ultimate objective, many instructional decisions are inherently subjective. Quantitative data serve as an objective means of evaluating whether such subjective approaches are effective, which is precisely the central aim of this study.

Effective teaching requires selecting appropriate instructional strategies to address students’ learning challenges. Although instructional adjustments inevitably involve both successes and failures, continuously refining teaching practices enables students to overcome learning bottlenecks while minimizing student attrition. Through systematic data analysis and evidence-based instructional decisions, I hope to continuously improve the effectiveness and consistency of my teaching practice.

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